

PIERCE LUU

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Education

University of Toronto

Expected Graduation: 2029

Bachelor of Applied Science in Computer Engineering

- **Relevant Coursework:** ECE244 Object Oriented Programming (C++), APS105 Programming with C

Technical Skills

Languages: Python, C, C++, JavaScript, TypeScript, HTML/CSS, SQL (PostgreSQL)

Web: FastAPI, Flask, Node.js, Express.js, React, Next.js

AI, Data & ML: LangChain, LangGraph, NumPy, pandas, scikit-learn, PyTorch

Cloud & DevOps: Git, GitHub Actions, Docker, Kubernetes, AWS (EC2, S3, Lambda, DynamoDB, API Gateway, ECR, EKS, CloudWatch, Bedrock)

Experience

University of Toronto

May 2026 – September 2026

Undergraduate Researcher and Software Developer

Toronto, Ontario

- Shipped a **Three.js** 3D virtual lab for BME safety training now used across **3 undergraduate BME courses**, saving **3 hours** of in-lab orientation time per student
- Built a branching scenario sequencer on top of the lab twin so students can practice hazard response interactively without live lab risk
- Engineered an on-app local LLM pipeline with **RAG** over course-specific rules and equipment context to draft scenarios from lab protocols, speeding generation by **40%** and saving **45+ minutes** of instructor hand authoring per scenario

Ontario Online Schools

Jan 2026 – May 2026

Back-End Developer Intern

Remote

- Developed backend services using **Node.js** and **Express.js** for the **Next.js** platform serving **40+ students**
- Built an AI chatbot using the **Anthropic API** and a high-level, reducing support inquiries by **~65%**
- Configured **GitHub Actions** with **20+** unit tests to replace manual QA and releases, cutting deployment time by **70%**
- Improved SEO across **9** pages by optimizing metadata and resolving Lighthouse issues, increasing SEO scores by **15%**

GamePill

Feb 2024 – June 2024

Game Development and Quality Assurance Intern

Concord, Ontario

- Refined gameplay systems debugging **10+ issues** and contributing to rapid prototypes in **Unreal Engine (C++)**.
- Documented **150+ bugs** and improvement suggestions, helping accelerate development iteration by **15%**

Projects

Traffer: Traffic Monitoring with MLOps Pipeline | *Docker, Kubernetes, PyTorch, FastAPI*

- Engineered a traffic detection pipeline serving a **PyTorch** object detection model via **FastAPI** on **Kubernetes** (Minikube), achieving sub-**200ms** inference and **60%** Docker image reduction through multi-stage builds
- Automated CI/CD running **40 Pytest** unit tests, cutting deployment time to **under 3** minutes
- Packaged MLOps utilities for metrics reporting, batch inference, and model versioning, achieving **95%+** code coverage

TeddyTalk: Emotion Detecting Arduino Powered Talking Robot | *Arduino, Python, OpenCV, Gemini, ElevenLabs*

- Developed a **Python** backend using **OpenCV** for face detection and fine-tuned a FER+ based emotion recognition model on data, achieving **95%+ accuracy** for real-time inference
- Generated and read messages out loud using Gemini and ElevenLabs
- Won Best Use of Arduino UNO Q and Best Use of ElevenLabs API at MakeUofT (Hackathon) out of **69 teams**

Silo Labs: Agentic Coding for Embedded Systems | *Python, FastAPI, LangGraph, Next.js, Claude, Docker*

- Built an AI firmware platform that turns natural-language hardware requests into pinouts, clock configs, compiled C drivers, and a live circuit simulation for the RP2040 using a 10-agent LangGraph pipeline
- Parallelized per-peripheral work across up to 6 concurrent feature branches with LangGraph Send and a custom state reducer, streaming live token, cost, and latency metrics over SSE
- Grounded generation on a knowledge base of **37** peripherals and **~1,000** registers so drivers use real hardware addresses